SEMESTER 1
2021-2022

CS370FZ
Computation and Complexity

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Time allowed: 2 hours

Answer all **three** questions

All three questions carry equal marks

Instructions

	Yes	No
Log Books Allowed	☐	✓☐
Formula Tables Allowed	☐	✓”☐
Other Allowed (*enter details)*	☐	✓”☐

General (*enter details)*

[25 marks]

1

(a) Give self-contained definitions of the following languages/sets: A_{TM} (the acceptance problem), $Halt_{TM}$ (the halting problem), and TM M decides Language L. [9 marks]

(b) Show that A_{TM} is recognizable (construct a recognizer for it). [4 marks]

(c) Show that A_{TM} is undecidable. (You are required to use the diagonalization technique) [8 marks]

(d) Is $\overline{A_{TM}}$ recognizable? Explain your answer. [4 marks]

[25 marks]

2

(a) Give the statement of Rice's Theorem. [5 marks]

(b) Prove Rice's Theorem (you can use the recursion theorem). [8 marks]

(c) Give the definitions of the following concepts: (1) Language A is mapping reducible to language B. (2) Language A is Turing reducible to language B. [6 marks]

(d) Show that $A_{TM} \leq_m HALT_{TM}$. [6 marks]

[25 marks]

3

(a) Give the definitions of the following classes: Class P and Class NP. [4 marks]

(b) Give the definitions of the following sets: PATH, SAT, and CLIQUE. Your definition should be self-contained (e.g. for CLIQUE, explain what a k-clique is). [6 marks]

(c) Show that CLIQUE is in NP (give a verifier or a nondeterministic polynomial time Turing machine that decides CLIQUE). [5 marks]

(d) Recall the proof 3SAT is polynomial time mapping reducible to VERTEX-COVER (3SAT$\leq_P$ VERTEX-COVER). From the following ϕ construct the associated graph G. Explain how to obtain an 8-vertex-cover for G from a satisfying assignment for ϕ. [10 marks]

$$\phi = (x_1 \vee x_1 \vee x_2) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_2 \vee x_2)$$